

Domain and Range

ANSWER KEY



Domain from equations

- 1 State the domain of each function:

a $f(x) = \frac{1}{x-4}$ All real x with $x \neq 4$.

b $g(x) = \sqrt{2x-6}$ $2x-6 \geq 0$, so $x \geq 3$.

c $h(x) = \frac{x+1}{x^2-9}$ All real x with $x \neq 3$ and $x \neq -3$.

d $k(x) = \sqrt[3]{x}$ All real numbers.

- 2 Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

We need $5-x \geq 0$, so $x \leq 5$, and $x \neq 1$. Domain: $(-\infty, 1) \cup (1, 5]$.

Domain and range from graphs

- 3 The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.

Domain: all real numbers. Range: $y \geq -2$.

- 4 The graph of $y = \sqrt{x+3}$ is shown. State the domain and range of the function.

Domain: $x \geq -3$. Range: $y \geq 0$.

Finding the range algebraically

- 5 By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

$f(x) = (x+2)^2 - 3$. Since $(x+2)^2 \geq 0$, the range is $y \geq -3$.

- 6 Consider $f(x) = -2(x-1)^2 + 5$.

a State the coordinates of the vertex. Vertex: $(1, 5)$.

b Does the parabola open upward or downward? Downward, since the leading coefficient $-2 < 0$.

c State the range of f . $y \leq 5$.